MA1522 Linear Algebra for Computing

Lesson 19: Linear Transformations

Announcements

- **Online Quiz 5** opens **Saturday 31 October, 0:00**, and is due **Sunday 1 November, 23:59**.

- This is the last online quiz. The best four of your five quiz scores count toward the final grade.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson19` (e.g. `A1234567Z_lesson19.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 19**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev



Section 7.1 Linear Transformation

Definition

A mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ is a **linear transformation** if for all vectors $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$ and scalars α, β,

$$T(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v}).$$

$\mathbb{R}^n$ is the **domain** and $\mathbb{R}^m$ is the **codomain** of T.

Equivalently, $T(\alpha\mathbf{u}) = \alpha T(\mathbf{u})$ and $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$ and scalars α. By induction,

$$T(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_k\mathbf{u}_k) = c_1 T(\mathbf{u}_1) + c_2 T(\mathbf{u}_2) + \cdots + c_k T(\mathbf{u}_k).$$

Section 7.1 Remarks

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation.

1. The values of T at $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ determine the values of T at $\text{span}(S)$.
   Proof: if $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$, then by linearity $T(\mathbf{v}) = c_1T(\mathbf{v}_1) + \cdots + c_kT(\mathbf{v}_k)$.

2. $T(\mathbf{0}) = \mathbf{0}$.
   Proof: $T(\mathbf{0}) = T(0 \cdot \mathbf{0}) = 0 \cdot T(\mathbf{0}) = \mathbf{0}$.

3. Every $m \times n$ matrix A defines a linear transformation $T_A : \mathbb{R}^n \to \mathbb{R}^m$ by $T_A(\mathbf{u}) = A\mathbf{u}$.
   Proof: $T_A(\alpha\mathbf{u} + \beta\mathbf{v}) = A(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha A\mathbf{u} + \beta A\mathbf{v} = \alpha T_A(\mathbf{u}) + \beta T_A(\mathbf{v})$.

Section 7.1 Standard Matrix

Theorem

A mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation if and only if there is a unique $m \times n$ matrix A such that $T(\mathbf{u}) = A\mathbf{u}$ for all $\mathbf{u}$ in $\mathbb{R}^n$. The matrix A is given by

$$A = \begin{pmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{pmatrix},$$

where $E = \{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is the standard basis for $\mathbb{R}^n$.

If T is a linear transformation, then by linearity

$$T\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = T(c_1\mathbf{e}_1 + \cdots + c_n\mathbf{e}_n) = c_1T(\mathbf{e}_1) + \cdots + c_nT(\mathbf{e}_n) = \begin{pmatrix} T(\mathbf{e}_1) & \cdots & T(\mathbf{e}_n) \end{pmatrix}\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}.$$

Definition

This unique matrix A is called the **standard matrix**, or matrix representation, of T.

Question 1

1. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation such that

$$T\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}2\\-5\end{pmatrix}, \qquad T\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}1\\4\end{pmatrix}, \qquad T\begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}7\\-3\end{pmatrix}.$$

   Find the standard matrix A of T.

2. Suppose the standard matrix of a linear transformation $T' : \mathbb{R}^2 \to \mathbb{R}^3$ is

$$B = \begin{pmatrix}3 & 1\\-2 & 7\\5 & -3\end{pmatrix}.$$

   Write down the formula for $T'\begin{pmatrix}x\\y\end{pmatrix}$.

Question 1 — solution: part 1

1. Write each standard basis vector as a combination of the given vectors:

$$\mathbf{e}_1 = \begin{pmatrix}1\\0\\0\end{pmatrix}, \qquad \mathbf{e}_2 = \begin{pmatrix}1\\1\\0\end{pmatrix} - \begin{pmatrix}1\\0\\0\end{pmatrix}, \qquad \mathbf{e}_3 = \begin{pmatrix}1\\1\\1\end{pmatrix} - \begin{pmatrix}1\\1\\0\end{pmatrix}.$$

By linearity,

$$T(\mathbf{e}_1) = \begin{pmatrix}2\\-5\end{pmatrix}, \quad T(\mathbf{e}_2) = \begin{pmatrix}1\\4\end{pmatrix} - \begin{pmatrix}2\\-5\end{pmatrix} = \begin{pmatrix}-1\\9\end{pmatrix}, \quad T(\mathbf{e}_3) = \begin{pmatrix}7\\-3\end{pmatrix} - \begin{pmatrix}1\\4\end{pmatrix} = \begin{pmatrix}6\\-7\end{pmatrix}.$$

$$A = \begin{pmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & T(\mathbf{e}_3) \end{pmatrix} = \begin{pmatrix} 2 & -1 & 6 \\ -5 & 9 & -7 \end{pmatrix}.$$

Check: $A(1,1,0)^T = (2-1,\ -5+9)^T = (1,4)^T$. ✓

Question 1 — solution: part 2

2. T' is multiplication by its standard matrix:

$$T'\begin{pmatrix} x \\ y \end{pmatrix} = B\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ -2 & 7 \\ 5 & -3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x + y \\ -2x + 7y \\ 5x - 3y \end{pmatrix}.$$

Check: $T'(\mathbf{e}_1) = (3, -2, 5)^T$, the first column of B. ✓ □

Challenge 1

Let $T_1 : \mathbb{R}^n \to \mathbb{R}^m$ and $T_2 : \mathbb{R}^m \to \mathbb{R}^p$ be two linear transformations. Define $T_2 \circ T_1 : \mathbb{R}^n \to \mathbb{R}^p$ by

$$(T_2 \circ T_1)(\mathbf{x}) = T_2\big(T_1(\mathbf{x})\big).$$

Let $[T]$ denote the standard matrix of T.

1. Let $n = 2$, $m = 3$, $p = 2$, with

$$[T_1] = \begin{pmatrix} a & b \\ c & d \\ e & f \end{pmatrix}, \qquad [T_2] = \begin{pmatrix} u & v & w \\ x & y & z \end{pmatrix}.$$

   Compute $(T_2 \circ T_1)\begin{pmatrix} s \\ t \end{pmatrix}$ directly.

2. Show that $T_2 \circ T_1$ is a linear transformation, and that

$$[T_2 \circ T_1] = [T_2]\,[T_1] \qquad \text{(matrix multiplication)}.$$

Section 7.1 Representation with Respect to a Basis

Definition

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation and $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ a basis for $\mathbb{R}^n$. The **representation of T with respect to S** is the $m \times n$ matrix

$$[T]_S = \begin{pmatrix} T(\mathbf{u}_1) & T(\mathbf{u}_2) & \cdots & T(\mathbf{u}_n) \end{pmatrix}.$$

Theorem

For every $\mathbf{v}$ in $\mathbb{R}^n$, $T(\mathbf{v}) = [T]_S[\mathbf{v}]_S$. Let $P_{E\to S}$ be the transition matrix from the standard basis E to S, and $P_{S\to E}$ the transition matrix from S to E. Then the standard matrix A of T satisfies

$$A = [T]_S\, P_{E\to S}, \qquad [T]_S = A\, P_{E\to S}^{-1} = A\, P_{S\to E}.$$

Exercise. Show that the transition matrix $P_{S\to E}$ is exactly $[I]_S$, where $I : \mathbb{R}^n \to \mathbb{R}^n$ is the linear transformation given by $I(\mathbf{x}) = \mathbf{x}$.

Question 2

Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear transformation such that

$$T\begin{pmatrix}1\\0\\3\end{pmatrix} = \begin{pmatrix}-7\\5\\-2\end{pmatrix}, \qquad T\begin{pmatrix}1\\-1\\3\end{pmatrix} = \begin{pmatrix}-6\\2\\-4\end{pmatrix}, \qquad T\begin{pmatrix}-1\\0\\-2\end{pmatrix} = \begin{pmatrix}4\\-3\\1\end{pmatrix}.$$

Let $S = \left\{ \begin{pmatrix}1\\0\\3\end{pmatrix}, \begin{pmatrix}1\\-1\\3\end{pmatrix}, \begin{pmatrix}-1\\0\\-2\end{pmatrix} \right\}$, a basis for $\mathbb{R}^3$.

1. Write down $[T]_S$, and find the transition matrix P from the standard basis E to S.

2. Find the standard matrix A of T, and a formula for $T\begin{pmatrix}x\\y\\z\end{pmatrix}$.

Question 2 — solution: part 1

1. The columns of $[T]_S$ are the given images:

$$[T]_S = \begin{pmatrix} -7 & -6 & 4 \\ 5 & 2 & -3 \\ -2 & -4 & 1 \end{pmatrix}.$$

P is the inverse of $U = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{pmatrix}$. Reduce $(U \mid I)$: $R_3 - 3R_1$ and $-R_2$, then $R_1 - R_2 + R_3$:

$$\left(\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ 3 & 3 & -2 & 0 & 0 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & 1 & 1 \\ 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array}\right),$$

$$P = U^{-1} = \begin{pmatrix} -2 & 1 & 1 \\ 0 & -1 & 0 \\ -3 & 0 & 1 \end{pmatrix}.$$

Question 2 — solution: part 2

2. Multiply, $A = [T]_S\, P$:

$$A = \begin{pmatrix} -7 & -6 & 4 \\ 5 & 2 & -3 \\ -2 & -4 & 1 \end{pmatrix} \begin{pmatrix} -2 & 1 & 1 \\ 0 & -1 & 0 \\ -3 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 & -3 \\ -1 & 3 & 2 \\ 1 & 2 & -1 \end{pmatrix}.$$

Check one column: $A\mathbf{u}_1 = (2-9,\, -1+6,\, 1-3)^T = (-7, 5, -2)^T = T(\mathbf{u}_1)$. ✓

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2x - y - 3z \\ -x + 3y + 2z \\ x + 2y - z \end{pmatrix}.$$

□

Question 3

Let $\mathbf{u}_1 = \begin{pmatrix} 1 \\ -3 \\ 3 \end{pmatrix}$, $\mathbf{u}_2 = \begin{pmatrix} -2 \\ 5 \\ 1 \end{pmatrix}$, $\mathbf{u}_3 = \begin{pmatrix} 0 \\ -1 \\ 7 \end{pmatrix}$, and let k be a real number. Suppose $T : \mathbb{R}^3 \to \mathbb{R}^2$ is a linear transformation with

$$T(\mathbf{u}_1) = \begin{pmatrix} 4 \\ -3 \end{pmatrix}, \qquad T(\mathbf{u}_2) = \begin{pmatrix} -1 \\ 5 \end{pmatrix}, \qquad T(\mathbf{u}_3) = \begin{pmatrix} 7 \\ k \end{pmatrix}.$$

1. Find the value of k for which such a linear transformation T exists.

2. For this value of k, find the standard matrix of every such T. *Hint:* $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{e}_3\}$ is a basis for $\mathbb{R}^3$; let $T(\mathbf{e}_3) = \begin{pmatrix} a \\ b \end{pmatrix}$.

Question 3 — solution: part 1

1. Look for a dependence among $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$:

$$\mathbf{u}_2 + 2\mathbf{u}_1 = \begin{pmatrix} -2+2 \\ 5-6 \\ 1+6 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 7 \end{pmatrix} = \mathbf{u}_3.$$

By linearity,

$$T(\mathbf{u}_3) = T(\mathbf{u}_2) + 2T(\mathbf{u}_1) = \begin{pmatrix} -1 \\ 5 \end{pmatrix} + 2\begin{pmatrix} 4 \\ -3 \end{pmatrix} = \begin{pmatrix} 7 \\ -1 \end{pmatrix}.$$

So T can exist only if $k = -1$; part 2 shows that for $k = -1$ it does.

Question 3 — solution: part 2

2. With $S' = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{e}_3\}$ and $W = \begin{pmatrix}\mathbf{u}_1 & \mathbf{u}_2 & \mathbf{e}_3\end{pmatrix}$, invert W: $R_2 + 3R_1$, $R_3 - 3R_1$; then $R_3 + 7R_2$, $-R_2$, $R_1 + 2R_2$:

$$\left(\begin{array}{ccc|ccc} 1 & -2 & 0 & 1 & 0 & 0 \\ -3 & 5 & 0 & 0 & 1 & 0 \\ 3 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|ccc} 1 & -2 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 3 & 1 & 0 \\ 0 & 7 & 1 & -3 & 0 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -5 & -2 & 0 \\ 0 & 1 & 0 & -3 & -1 & 0 \\ 0 & 0 & 1 & 18 & 7 & 1 \end{array}\right).$$

Then $A = [T]_{S'}\, W^{-1}$:

$$A = \begin{pmatrix} 4 & -1 & a \\ -3 & 5 & b \end{pmatrix} \begin{pmatrix} -5 & -2 & 0 \\ -3 & -1 & 0 \\ 18 & 7 & 1 \end{pmatrix} = \begin{pmatrix} 18a - 17 & 7a - 7 & a \\ 18b & 7b + 1 & b \end{pmatrix}, \qquad a, b \in \mathbb{R}.$$

Each choice of a, b gives a different T, so T is not unique. □

Challenge 2

Fix a subset $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ of vectors in $\mathbb{R}^n$. Let $S' = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be an arbitrary set of vectors in $\mathbb{R}^m$. Prove the following.

1. If $\text{span}(S) = \mathbb{R}^n$, then there is at most one linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ such that $T(\mathbf{v}_i) = \mathbf{u}_i$ for $i = 1, \ldots, k$.

2. If S is linearly independent, then there is at least one linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ such that $T(\mathbf{v}_i) = \mathbf{u}_i$ for $i = 1, \ldots, k$.

Next lesson

Next lesson (**Tue 3 Nov**, Sections 7.1 & 7.2), I will focus on

1. the range and the kernel of a linear transformation, and
2. injective linear transformations.

Please watch the video clips beforehand.